Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/02/2025
PI Name:	Risca, Viviana Ph.D.		
Project Title:	Epigenetic State Modulation of Radiation-Induced DNA Damage: Nanoscale Modeling and Validation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
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Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:	Dr. Antony Jeevarajan requested that Dr. Karen Pickering be added as the institutional Co-Investigator.		
COI Name (Institution):	Plante, Ianik Ph.D. (NASA Johnson Space Center) Jeevarajan, Antony Ph.D. (NASA Johnson Space Center) Pickering, Karen (NASA Johnson Space Center)		
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Task Description:

BACKGROUND

The risks of cellular dysfunction associated with exposure to space radiation, including transcriptional and epigenetic perturbations and genomic instability due to DNA breaks, have been studied in cell lines, with DNA repair foci and products as the main readouts. Such genetic and cell biological readouts show that high linear energy transfer (LET) charged nuclei, such as those found in galactic cosmic rays (GCR), cause persistent cellular changes in stress response and genomic integrity. These effects are different from the effects of low-linear energy transfer (LET) radiation such as X-rays and occur in the context of the genome-wide epigenetic landscape of each cell, which includes nucleosome positions, nucleosome modifications, and variant histone substitutions in those nucleosomes. Epigenetic states differ in chromatin fiber conformations, with transcriptionally active chromatin adopting more open, extended structures. These differences can affect DNA break patterns in response to ionizing radiation, potentially creating distinct DNA repair and signaling outcomes. The epigenetic state landscape of a cell depends on its differentiation state, cell type, and responses to external stimuli. Because it is not practical to experimentally investigate every cell type, a more generalizable approach is needed to predict how the cell's distinctive epigenetic landscape will interact with radiation to give rise to a certain pattern of DNA breaks and associated cellular response. A generalizable approach that takes local epigenetic map information into account can leverage the large and diverse epigenomic data sets available for a large number of human cell types. Previous investigations of chromatin structure's role in regulating DNA damage by radiation assumed that chromatin adopts stable, regular structures such as 30-nm fibers. Recently emerging consensus in the field suggests this single-structure view is inaccurate and the ensemble of conformational fluctuations of the fiber must be taken into account.

HYPOTHESIS

We hypothesize that the pattern and lethality of DNA breaks generated at a given genomic locus depend on the combination of (1) the incoming ionizing radiation, with differences between low LET photons and high LET GCRs, and (2) the epigenetic state of that locus, which is associated with a characteristic ensemble of chromatin fiber conformations.

DELIVERABLES

We propose to develop a generalizable mechanistic approach to determining how DNA breaks are generated by ionizing radiation including GCRs and photons. We will integrate realistic chromatin fiber ensembles with Monte Carlo simulations of photons or GCR nuclei interacting with those fibers, and Green's function based calculation of radiochemistry kinetics after the particle delivers its energy. The chromatin fiber ensembles will be generated through a coarse-grained simulation based on a stretchable shearable worm-like chain model of linker DNA between nucleosomes that is constrained by pairwise DNA-DNA contact data. We will measure chromatin fiber contact distances, simulate sub-kilobase chromatin conformation ensembles consistent with those contact distances, and predict how these ensembles give rise to ensembles of DNA break patterns. These measurements and simulations will be carried out for multiple chromatin states across several cell types as well as for in vitro reconstituted chromatin fibers in order to build a general, cell type independent model of the relationship between epigenetic state and vulnerability to radiation induced DNA damage. The resulting software package will enable the simulation of user-programmable chromatin states, to produce chromatin state specific predictions of expected DNA fragmentation patterns for each type of heavy ion or photon of incoming radiation. These fragmentation patterns can then form the basis for future mechanistic studies of the cell's differential repair and signaling responses to varied break cluster types.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our research develops technology for mapping DNA breaks caused by radiation onto the human genome and studies how the sensitivity to radiation varies across the human genome. Our data and methods will have direct applicability to determining cell type specific sensitivity to radiation therapy used to treat many cancers. We anticipate that our results will also aid cancer prevention here on Earth in addition to helping to advance our understanding of the health risks associated with space travel.

Task Progress:

In the last reporting period, our overall goal has been to integrate our simulation work with the experiment.
We have developed a machine-learning approach that draws on our coarse-grained simulations of chromatin structures to train a convolutional neural network. We use this neural network to predict chromatin structure parameters from RICC-seq data.

Continuing to work with the version of RITRACKS described in the last reporting period, we have applied RITRACKS to simulations with multiple nucleosomes with the goal of understanding how the compaction of nucleosome fibers affects the protection of DNA by chromatin. These simulations have proven difficult because they are very computationally intensive, but we have successfully obtained results in which we reproduce the nucleosome entry-exit correlated cleavage events we see in RICC-seq experiments.

On the experimental side, we have adapted two protocols for use with cells irradiated with ionizing radiation: END-seq, which maps DNA double-strand breaks (DSBs), and GLOE-seq, which maps DNA single-strand breaks (SSBs), onto genomic DNA coordinates. These methods use single-ended mapping of breaks and do not depend on regions of high break density or on spatially correlated breaks to generate DNA fragments that can be sequenced, in contrast to RICC-seq, the method we were using in the prior reporting period.

We have performed 200 Gy gamma ray and Fe ion irradiations at NASA Space Radiation Laboratory (NSRL) on four cell types: K562 leukemia cells, BJ fibroblasts, IMR90 fibroblasts, and RPE-1 retinal pigment epithelial cells. Over the last reporting period, we have been processing banked samples from these experiments using END-seq and GLOE-seq, prioritizing the generation of data that reports on DNA break densities.

We have obtained quality control data that shows promisingly low levels of background breaks in un-irradiated cells. Preliminary analyses of the END-seq data in K562 cells show enrichment of breaks near promoters of genes, which is consistent with our prior results from X-ray experiments that indicated less protection of DNA by chromatin near de-compacted regulatory DNA regions like promoters and enhancers.

Bibliography Type:	Description: (Last Updated: 03/12/2025)
Abstracts for Journals and Proceedings	Plante I, Canaj H, Scortea A, Clerkin AB, Rendleman J, West DW, Weeks J, Risca VI. "Simulation of radiation-induced DNA damage with the code RITRACKS." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Canaj H, Scortea A, Rendleman J, West DW, Plante I, Risca VI. "Mapping DNA damage propensity by ionizing radiation with respect to epigenetic states." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Other Journals or Periodicals	Clerkin AB, Pagane N, West DW, Spakowitz AJ, Risca VI. "Determining mesoscale chromatin structure parameters from spatially correlated cleavage data using a coarse-grained oligonucleosome model." bioRxiv preprint server. Posted July 29, 2024. https://doi.org/10.1101/2024.07.28.605011 , Jul-2024
Articles in Peer-reviewed Journals	Plante I, West DW, Weeks J, Risca VI. "Simulation of radiation-induced DNA damage and protection by histones using the code RITRACKS." BioTech. 2024 Jun 5;13(2):17. https://doi.org/10.3390/biotech13020017 ; PubMed PMID: 38921049; PubMed Central PMCID: PMC11201919 , Jun-2024